

Molecular structure and transport of ionic liquid confined in asymmetric graphene-coated silica nanochannel

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ABSTRACT

The transport of ionic liquids (ILs) in nanochannel is greatly restricted, and facilitating transport of ILs is of great significance to their applications in gas separation, battery and supercapacitor, etc. In this work, we use molecular dynamic simulation to investigate the molecular structure, charge distribution, diffusion behavior and flow characteristics of [EMIM][BF₄] confined in asymmetric graphene-coated silica nanochannel to provide information for guiding the fabricate of nanodevice, and analyze the influence of nanochannel height, coating graphene and driving pressure. Water also studied as a comparison. The results indicate that [EMIM][BF₄] and water confined in silica nanochannel present layered structure, and coating graphene on the wall surface of silica nanochannel can enhance the layering effect, and uniformize the charge distribution and reduce the diffusion coefficient. [EMIM][BF₄] presents different flow states at varying nanochannel height, including Poiseuille flow and partial plunger flow, but water only presents Poiseuille flow. Coating graphene on wall surface of silica nanochannel results obvious positive slippage on the wall surface and the fast transport in nanochannel.

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1. Introduction

Ionic liquids (ILs) which is a new type of environmentally friendly solvent being liquid at room temperature with many unique properties, such as thermal stability, good solubility, non-volatile, designability, etc., has huge potential of application in many fields like gas separation, catalyst, battery and supercapacitor. Meanwhile, nanochannel whose dimension is less than 100 nm, not only has large specific surface area, but also induces the unique property of fluid confined in it which can't be observed from bulk fluid. In view of the great value of nanochannels in improving ILs application, theoretical simulations [1] and experimental techniques (such as Raman [2–4] and NMR spectroscopy [5], neutron and Rayleigh wing scattering [6–8], and X-ray and neutron diffraction [9,10]) are carried out to investigate the structure and properties of ILs under confinement. Futamura et al. found that the Coulomb order of ILs confined in nanopores were broken near the wall through X-ray scattering experiments [11]. Feng et al. [12] and Shim et al. [13] conducted MD simulations on the ion distribution confined in nanopores of different sizes. Jiang [14] studied the transport and structure of ILs in conical nanopores

under an external electric field, and found the charging performance of ILs in supercapacitors can be significantly improved at higher voltage.

Understanding the fluid flow behavior in nanochannel which is greatly affected by the nanochannel surface is benefitting to achieve fast transport of fluid [15], especially for ILs who have high viscosity. Jing and Bhushan studied the effect of boundary slip and electrical double layer induced by surface charge on the volumetric flow rate [16]. Wang et al. studied the influence of the width and chemical properties of graphene nanochannel on the structure and flow properties of ILs [17].

Graphene is verified to be a good material for increasing the fluid flow velocity by weakening the interaction between liquid and nanochannel surface. The nanodevice is commonly made of silica [18], but there is no study on the structure and transport behavior of ILs in nanochannels made of silica coated with graphene, especially the heterogeneous nanochannels, which is meaningful information for the design and fabrication of graphene-coated nanochannels. In addition, how the structure and transport behavior of ILs different from traditional molecular liquid is unclear. From the perspective of practical application, to get knowledge for designing reasonable nanochannel and controlling the fluid transport, the structure and flow properties of 1-ethyl-3-methylimidazolium tetrafluoroborate ([EMIM][BF₄]) confined in silica nanochannels with or without graphene coated is studied

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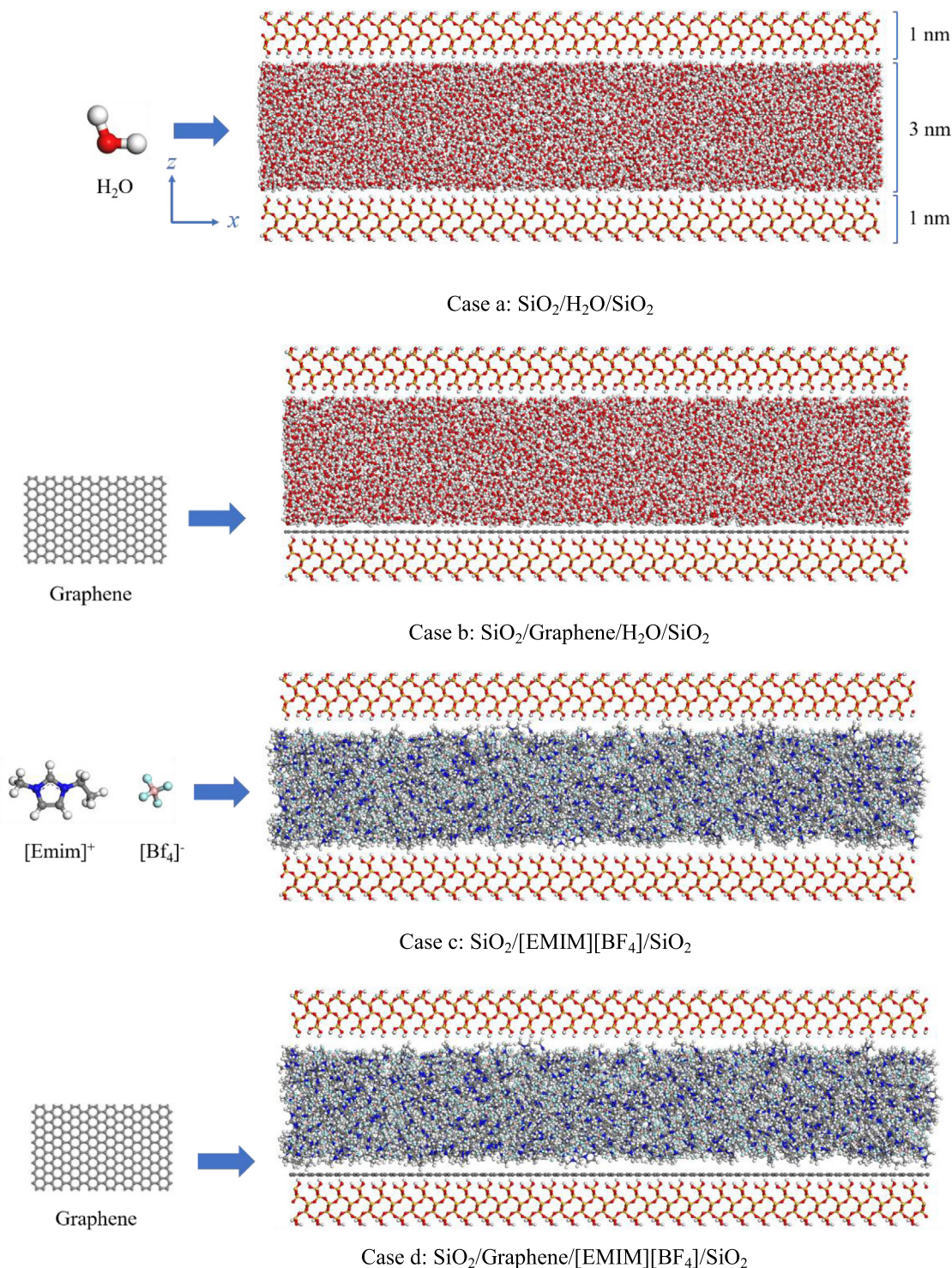


Fig. 1. The MD simulation model. a) Water confined in 3 nm silica nanochannel. b) Water confined in 3 nm nanochannel with monolayer coated-graphene. c) $[\text{EMIM}][\text{BF}_4]$ confined in 3 nm silica nanochannel. d) $[\text{EMIM}][\text{BF}_4]$ confined in 3 nm nanochannel with monolayer coated-graphene.

by molecular dynamics (MD) simulations. The influence of the height of nanochannel and coating graphene on interfacial structure of confined $[\text{EMIM}][\text{BF}_4]$ and nanofluidic flow behavior are focused on and compared with confined water. The self-diffusion coefficient of $[\text{EMIM}][\text{BF}_4]$ and water in nanochannels are also studied.

2. Simulation section

The nanochannel model is constructed by two parallel 1.0 nm silica substrates with varying height $d = 1.0$ to 5.0 nm, where the inner surface contains monolayer coated-graphene or not, as shown in Fig. 1. The silica substrate is composed of α cristobalite

and all silicon atoms that are incomplete tetrahedron are removed, while non-bridging oxygen atoms (only combined with one silicon atom) are saturated by hydrogen atoms [1,19]. The length of the channel along the x-axis direction is 14.9 nm, and the width along the y-axis direction is 5.0 nm.

We perform all the simulations by using large-scale atomic/molecular massively parallel simulation (LAMMPS) [20]. The all-atom optimized potential for liquid simulation (OPLS-AA) is used

for grapheme [21]. The SPC/E model is employed for water with SHAKE algorithm used to maintain rigidity of water molecules, which predicts density, viscosity and diffusivity accurately and is widely adopted in MD simulation of water transport [1,22]. The Amber force flied is used to describe the non-bonded interactions (including electrostatic and van der Walls interactions), bond, angle, and dihedrals in IL [23] and several parameters are obtained in the literature [24]. We adopt CLAYFF force flied in SiO₂, with

Table 1
Parameters of graphene, silica and water for the potential model in MD simulation.

Model	Atom	ε (kcal/mol)	σ (nm)	q (e)	Bond length (nm)	Angle bend (deg)
Graphene	C	0.06831	0.33997	0	$l_{C-C} = 0.142$	$\theta_{C-C-C} = 120$
	Si	0.00000184	0.33020	+2.1	$l_{Oh-H} = 0.1$	$\theta_{Si-Oh-H} = 109.47$
Silica	O	0.1554	0.3166	-1.05		
	Oh	0.1554	0.3166	-0.95		
	H	0	0	+0.425		
Water	O*	0.15535	0.3166	-0.8472	$l_{O^*-H^*} = 0.1$	$\theta_{H^*-O^*-H^*} = 109.47$
	H*	0	0	+0.4236		

Oh represents the non-bridged oxygen atom.
* represents atom in water.

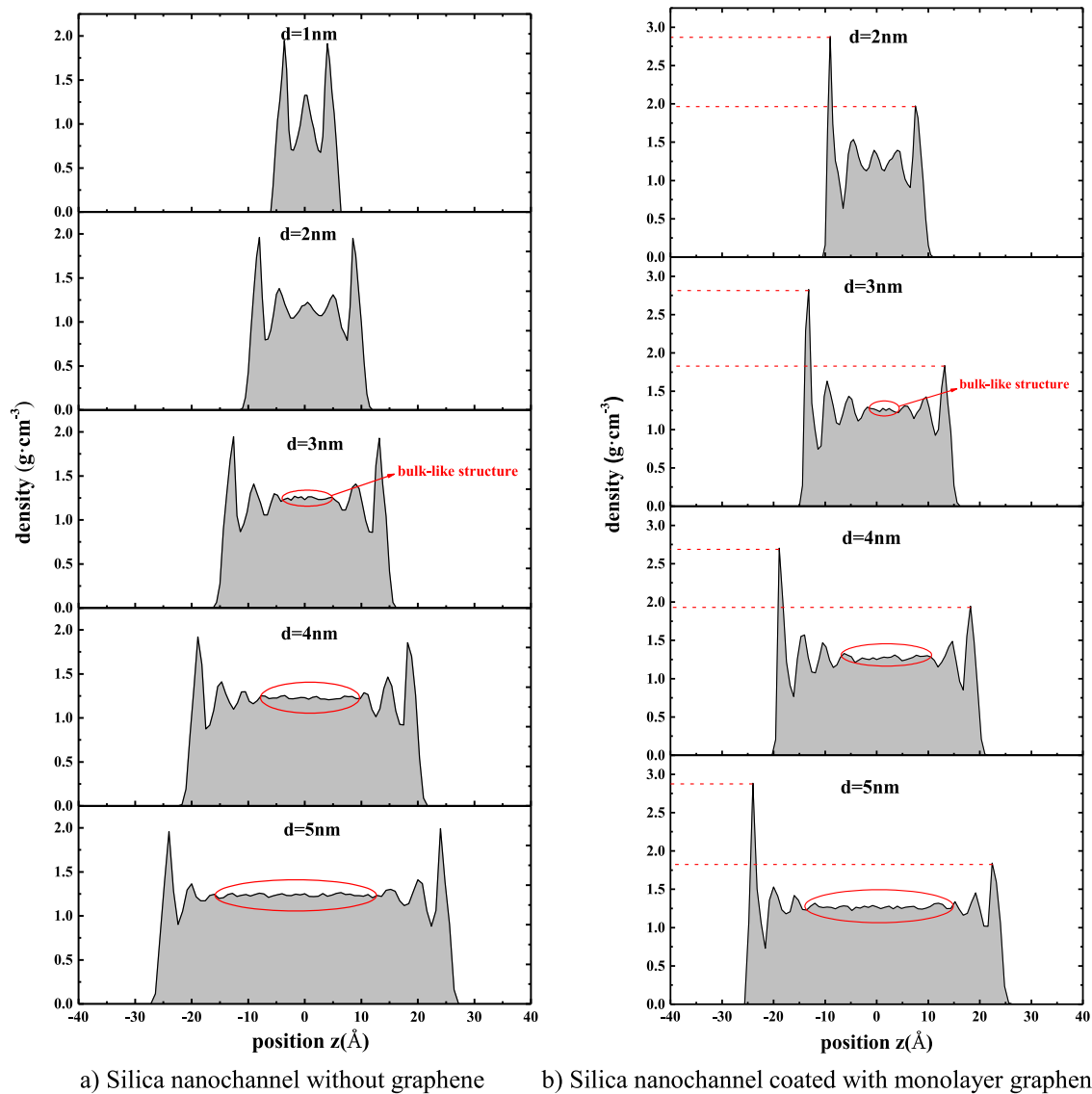


Fig. 2. ILs density distribution in different nanochannels (in z direction).

parameters shown in Table 1, which has predicted water properties in silica nanopores [1]. For non-bonded interactions between SiO_2 , graphene and water or IL, the van der Waals interactions is described in the Lennard-Jones 12-6 form with a cutoff of 1.2 nm. The long-range Coulombic interaction is computed using the particle-particle particle-mesh (PPPM) algorithm. The Lorentz-Berthelot mixing rules expressed as the Eq. (1) and Eq. (2)

are used to obtain the parameters between SiO_2 , graphene and water or IL.

$$\epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j} \quad (1)$$

$$\sigma_{ij} = \frac{\sigma_i + \sigma_j}{2} \quad (2)$$

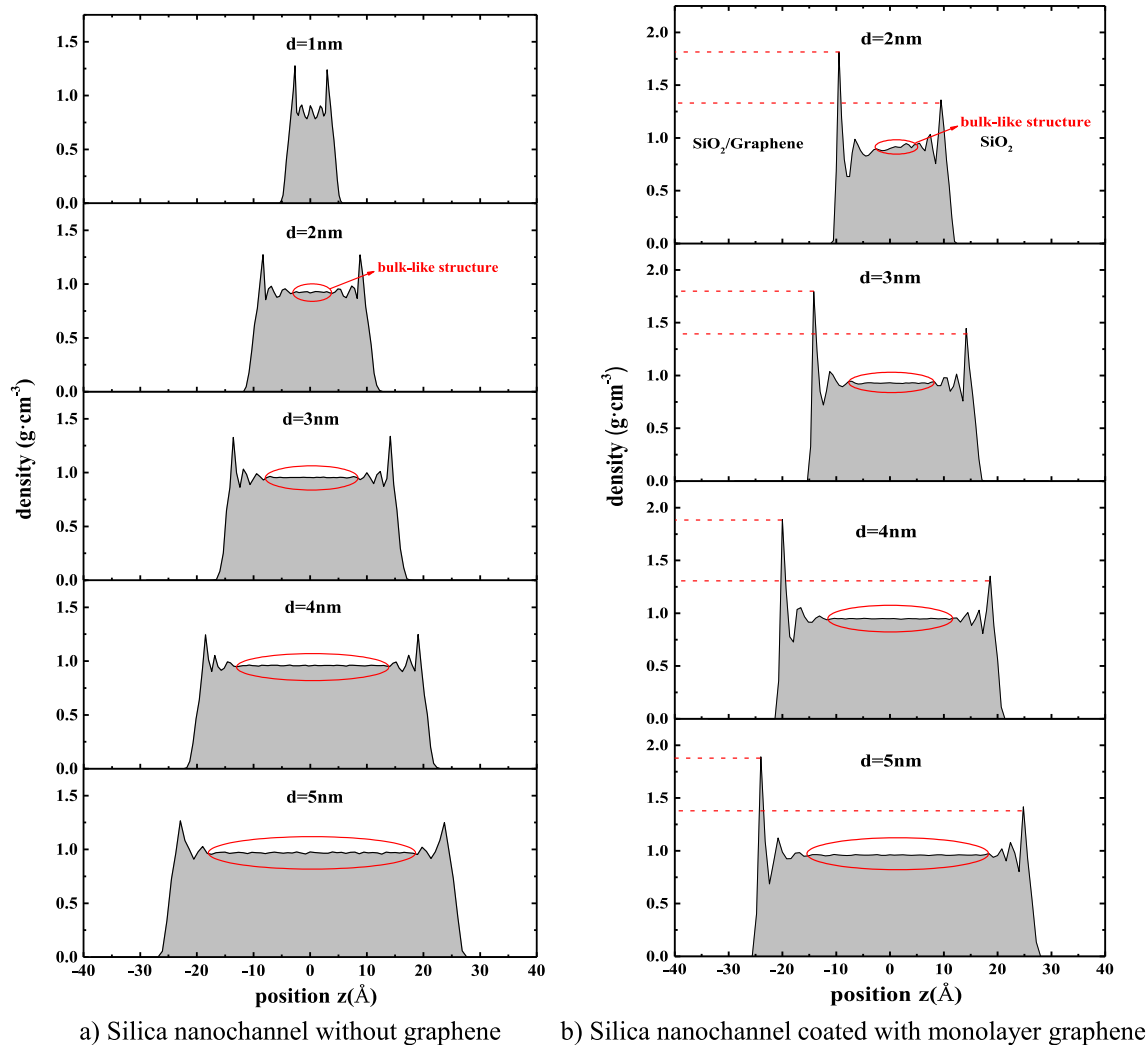


Fig. 3. Water density distribution in different nanochannels (in z direction).

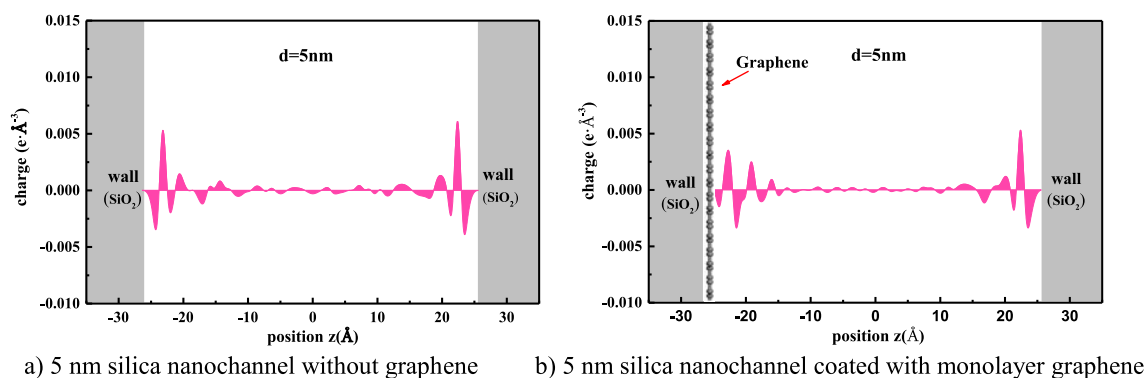


Fig. 4. The charge distribution of [EMIM][BF₄] in nanochannel.

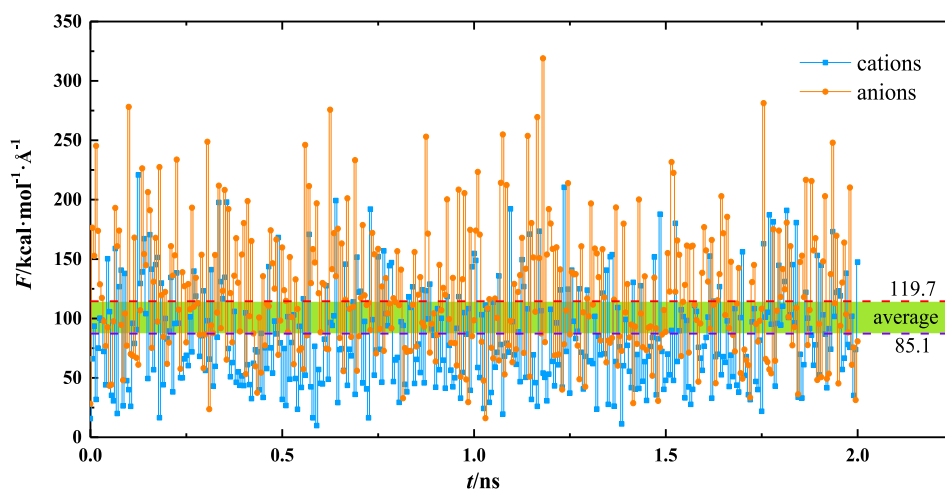


Fig. 5. Force between SiO₂ nanochannel wall and anions or cations.

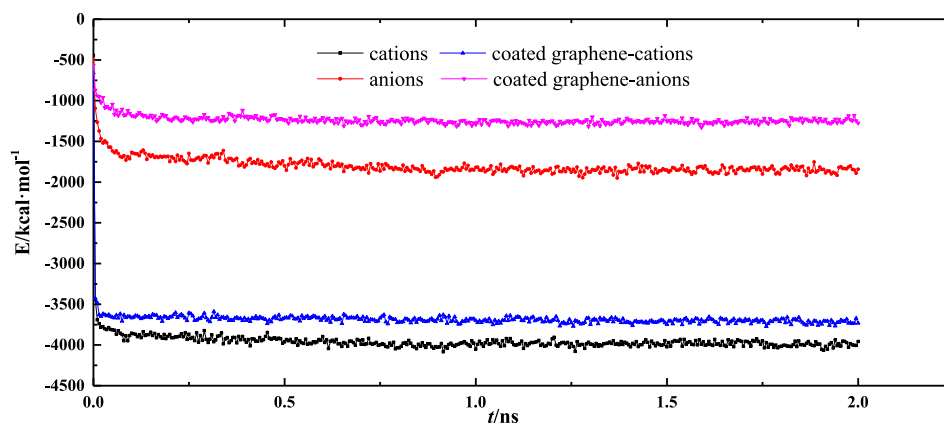


Fig. 6. Energy between SiO₂ nanochannel wall and anions or cations.

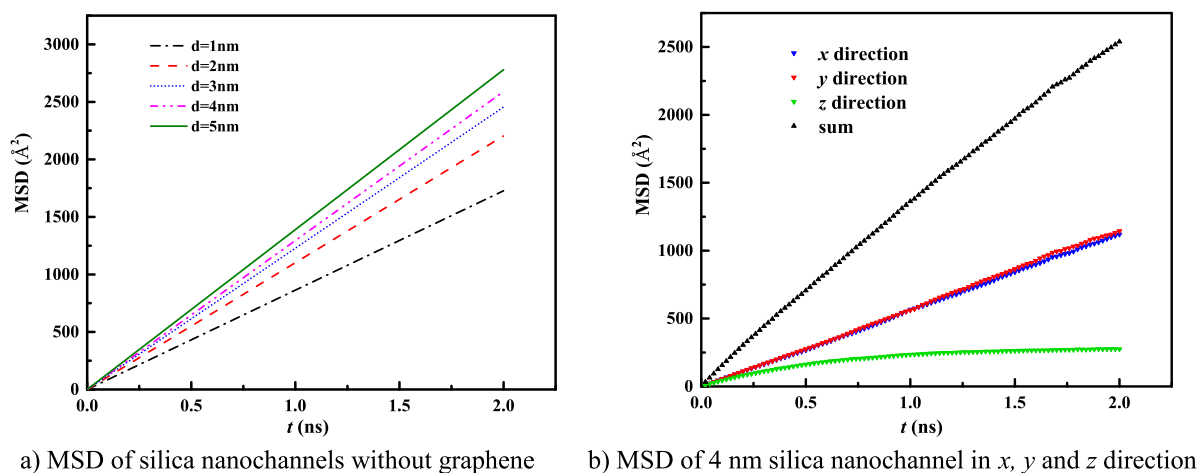


Fig. 7. The water molecules MSD in different nanochannels. The “sum” means the MSD in the entire space.

The parameters of interaction between the oxygen atoms in water and the carbon atoms in graphene are $\epsilon_{C-O} = 0.0936$ kcal/mol and $\sigma_{C-O} = 0.319$ nm, which predict a water contact angle (WCA) of 98.4° on graphene, consistent with experimental value [25]. The timestep for integrating Newton's equations of motion is 1.0 fs. We adopt periodic boundary conditions in both x and y direction while apply an open boundary condition in z direction.

Si, O, C atoms in silica and graphene are fixed during simulations. Water molecules and ILs are kept at 300 K using the Berendsen thermostat.

During the MD simulation, the system is first equilibrated in NVT ensemble for 4 ns. Then density distribution and mean square displacement (MSD) of water or ILs are calculated for 2 ns. After that, we apply an external force on every atom in water or ILs

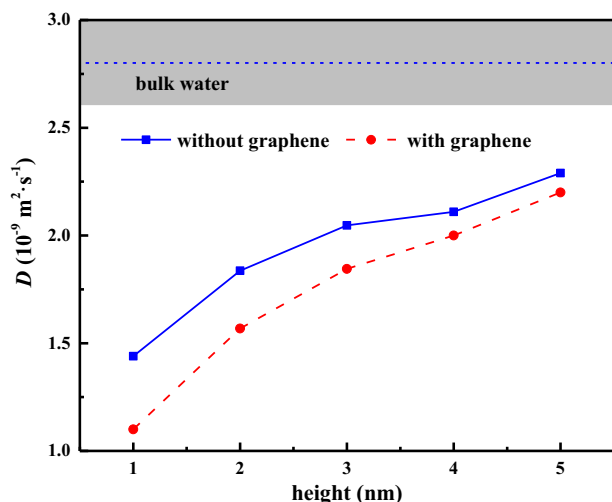


Fig. 8. The diffusion coefficient of water molecules in different nanochannels.

molecules in x direction to generate a driven flow. Fluid will reach a steady flow after about 6 ns relaxation and the data for subsequent analysis will be collected after extra 2 ns.

In order to verify the accuracy of simulation method in this study, the densities of water and [EMIM][BF₄] at 300 K and 0.1 MPa were calculated and compared with the experimental data reported by Wagner et al. [26] and Su et al. [27], while the absolute relative deviations are 1.1% and 3.1%, respectively.

3. Results and discussion

3.1. Molecular structure

The density distribution of [EMIM][BF₄] in silica nanochannel coated with or without monolayer graphene is shown in Fig. 2. From Fig. 2a, it can be clearly seen that due to the symmetry of nanochannel wall, the density distribution of [EMIM][BF₄] is also symmetry, and [EMIM][BF₄] presents a layered structure and forms dense layer near the wall surface. When the height of nanochannel is 1 nm and 2 nm where the confinement is strong, [EMIM][BF₄] presents a bilayer structure. As the height of the nanochannel increases, the confinement gradually

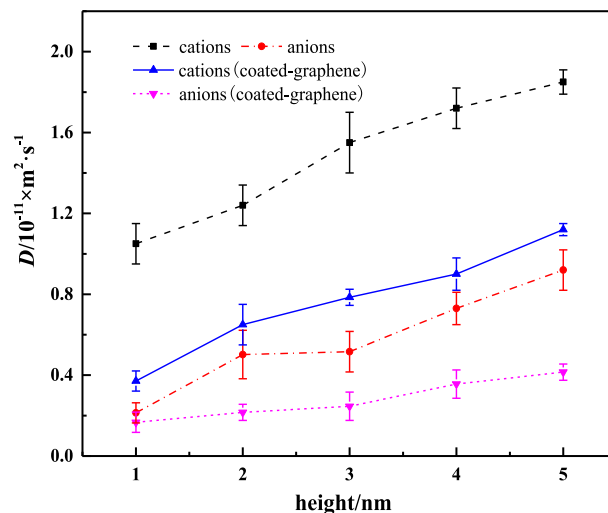


Fig. 10. Diffusion coefficient of anion and cation in nanochannels. The error bar represents the standard deviation of diffusion coefficient.

becomes weak and bulk-like structure appears. Confined water presents similar layered structure as shown in Fig. 3, but bulk-like structure appears when height of nanochannel is or larger than 2 nm. The density of [EMIM][BF₄] near wall is about 50% larger than bulk-like structure density, while the density of water near wall is about 30% larger than bulk-like density. This indicates that the layered structure of water is more affected by the wall surface of silica nanochannel than [EMIM][BF₄]. The reason is that the hydrogen bond network in the confined water plays an important role, while the hydroxyl groups on the wall surface of silica nanochannel destroy the hydrogen bonds in the dense water near the wall, and weakens the layered phenomenon. In [EMIM][BF₄], the electrostatic force between anion and cation is dominant, the layered phenomenon of [EMIM][BF₄] is less affected by the hydroxyl groups on the wall surface. When the nanochannel coated with monolayer graphene, the layered structure of [EMIM][BF₄] and water is more obvious and the density near the wall coated monolayer graphene is higher than that near the wall without coated monolayer graphene as shown in Fig. 2(b) and (b). It can be explained by that the graphene weakens the influence of hydroxyl groups on the

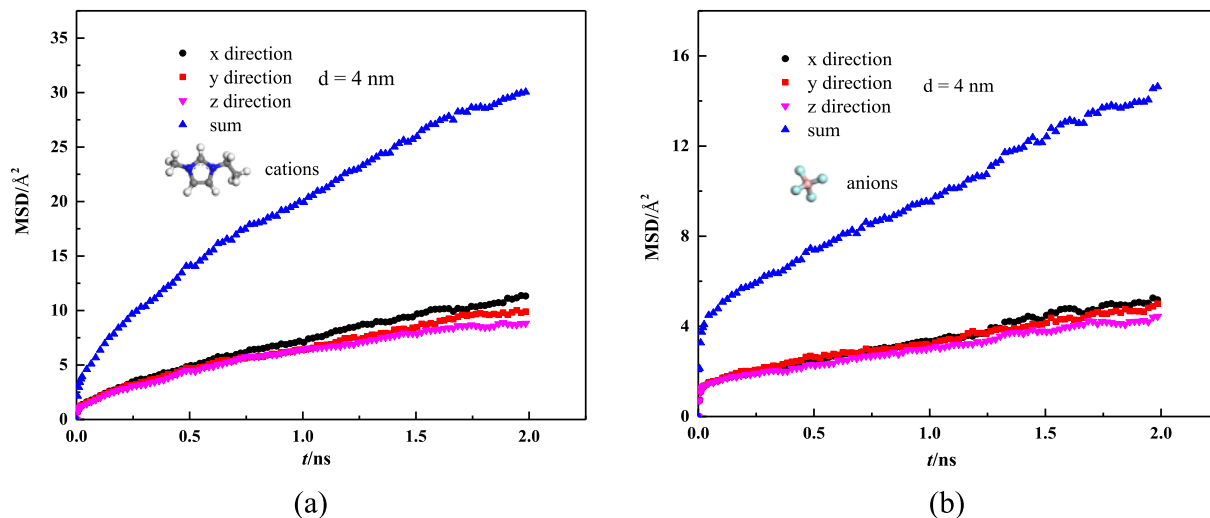


Fig. 9. MSD of cations and anions in different directions with the simulation time. a) cations b) anions.

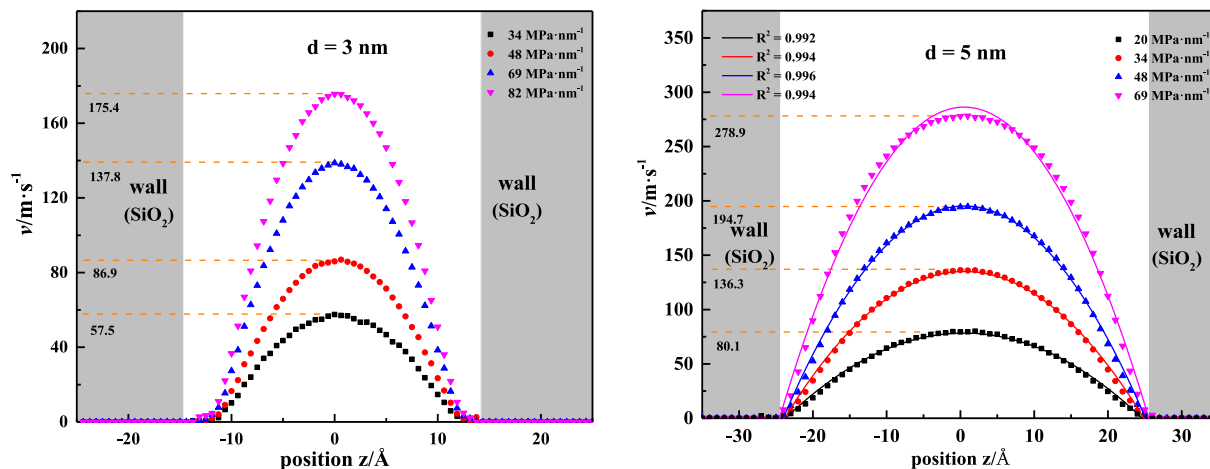


Fig. 11. Velocity profile of water confined in nanochannel under different driving pressure. The line represents the quadratic function of velocity with respect to nanochannel height in z direction.

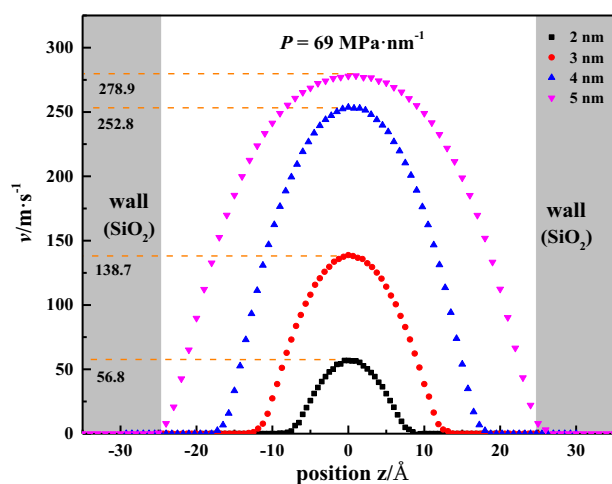


Fig. 12. Velocity profile of water confined in different nanochannels.

wall surface of silica nanochannel on the hydrogen bond in [EMIM][BF₄] and water.

The charge distribution of IL in nanochannel is important for ion transport due to its ionic nature. As shown in Fig. 4, we can see that there are charge layers of [EMIM][BF₄] near the wall regardless of whether the wall is coated with graphene. Far away from the wall, wall has no significant effect on Coulombic ordering, and the bulk-like charge distribution appears. The reason is that the Coulombic ordering of [EMIM][BF₄] is destroyed close to the wall surface due to that the wall has different interaction with the anion and cation. To prove that, the interaction force and energy between anions or cations and wall in 5 nm channel are simulated as shown in Fig. 5 and Fig. 6. It can be seen that the interaction force between wall and anions is stronger than that between wall and cations, which are 109.3 kcal·mol⁻¹·Å⁻¹ and 85.1 kcal·mol⁻¹·Å⁻¹, respectively. When one side of nanochannel is coated with monolayer graphene, as shown in Fig. 4b, the first peak and the second peak near that wall are lower than those near another wall, while the third peak and the fourth peak are higher than those near another wall. It can be concluded that: graphene makes the charge distribution more uniform, but result larger interaction energy of anion and cation with the wall which is also reflected by the more obvious layering effect.

3.2. Diffusion coefficient

This paper first calculates the MSD of water molecules in different nanochannels, as shown in Fig. 7. MSD curve reflects the movement of molecules over time. According to the Einstein diffusion relation [28], the diffusion coefficient of molecules can be calculated by

$$D = \lim_{t \rightarrow \infty} \frac{1}{2Nnt} \left\langle \sum_{j=1}^N (r_j(t) - r_j(0))^2 \right\rangle = \frac{1}{2n} \lim_{t \rightarrow \infty} \frac{d\text{MSD}(t)}{dt} \quad (3)$$

where D is the diffusion coefficient, N is the number of molecules in the system, and n is dimension of the studied system.

It can be seen from Fig. 7a that the MSD of water molecules in different nanochannels is linearly distributed with time. Nanochannel with smaller height has a stronger restriction to the free movement of water. Therefore, the growth rate of MSD curves and diffusion coefficient over time increase with the height of nanochannel as shown in Figs. 7 and 8. When the nanochannel coated with graphene, the diffusion coefficient in nanochannel with the same height is significantly reduced as shown in Fig. 8, this is because coated-graphene makes the layering of water more obvious, and the structure of water near the wall is more orderly, and the free movement of water molecules is more strongly restricted. The layered structure of water in the nanochannel greatly inhibits the diffusion behavior of water molecules, but the effect of graphene decreases with the increase of nanochannel height.

We also calculate MSD of water molecules in different directions, as shown in Fig. 7b. It can be seen that the MSD curve in x and y directions are linearly increasing over time, while the MSD in z direction tends to be stable over time due to the limitation of the nanochannel, so the restriction of the nanochannel on water is mainly in the z direction.

[EMIM][BF₄] confined in nanochannel shows different diffusion characteristics from water. Fig. 9 shows the MSD of anions and cations in the nanochannel with a height of 4 nm in different directions over the simulation time. It can be seen that the MSD is increasing with time, due to the restriction of [EMIM][BF₄] from the nanochannel, the MSD in z direction is smaller than that in x and y direction, but this gap is not as large as the MSD of water. The reason should be that the much greater viscosity of [EMIM][BF₄] than water makes it move slowly, the confinement of nanochannel on [EMIM][BF₄] in z direction is not significant as the on water.

Fig. 10 shows the variation of the diffusion coefficients of anion and cation with the height of nanochannel. The error bar represents the standard deviation of diffusion coefficient which is less than $2 \cdot 10^{-12} \text{ m}^2 \cdot \text{s}^{-1}$. It can be seen that the diffusion coefficients increase as the height increases regardless of the wall is coated with or without graphene. In the same nanochannel, the diffusion coefficient of cation is larger than that of anion due to the formation of O-H-F hydrogen bonds between the F atoms in anions and the hydroxyl groups on the wall surface slow the movement of anions. When the nanochannel is coated with graphene, the diffusion coefficients of cation and anion and their difference are all reduced. The reason is that, graphene breaks the O-H-F hydrogen bonds between anions and wall surface, and enhances the wall restrictions on anions and cations.

3.3. Flow behavior

1) Flow behavior of water

The flow behavior of water in nanochannel is essentially different from those of macroscopic water. First, velocity profile of water in nanochannel with a height of 3 nm is shown in Fig. 11. It can be seen that the velocity distribution of water changes parabolically with position in z direction as a Poiseuille flow. The velocity increases with the increase of driving pressure gradient, but water near wall presents a negative slippage which doesn't disappear with the increasing driving pressure gradient. This is because the viscous force inside water is smaller than the interaction force between water and wall. The flow velocity curve of water in

5 nm nanochannel under different driving pressure gradient is shown in Fig. 11. It can be seen that the flow is still a Poiseuille flow but no slippage, which also reflects that confinement of nanochannel is weak at high height.

In addition, the effect of the height of nanochannel on the water flow velocity under the same pressure gradient is shown in Fig. 12. It can be seen that under the same pressure gradient of $69 \text{ MPa} \cdot \text{nm}^{-1}$, when the height of nanochannel increases from 2 nm to 4 nm, the maximum water flow velocity increases from $56.8 \text{ m} \cdot \text{s}^{-1}$ to $252.8 \text{ m} \cdot \text{s}^{-1}$. Therefore, increasing the height of nanochannel gives an obvious increase in water flow velocity.

2) Flow behavior of [EMIM][BF₄]

The flow behavior of [EMIM][BF₄] confined in silica nanochannel with various heights 2–5 nm and coated with monolayer graphene are shown in Fig. 13. From Fig. 13, we can see that when the confinement is strong ($d = 2 \text{ nm}$), the velocity profile of [EMIM][BF₄] can be described by quadratic parabolic function and presents a Poiseuille flow. With the nanochannel height increases ($d \geq 3 \text{ nm}$), the confinement on [EMIM][BF₄] becomes weaker and the flow is no longer Poiseuille flow, but a partial plunger flow, which is consistent with the description in the literature that studied the flow behavior of [EMIM][BF₄] confined in different hydroxylated graphene nanochannel [17]. In addition, different from water, [EMIM][BF₄] shows positive slippage in all cases; as the channel height increase, the slip length increases instead of decreasing; the slip length also increases with the increasing pres-

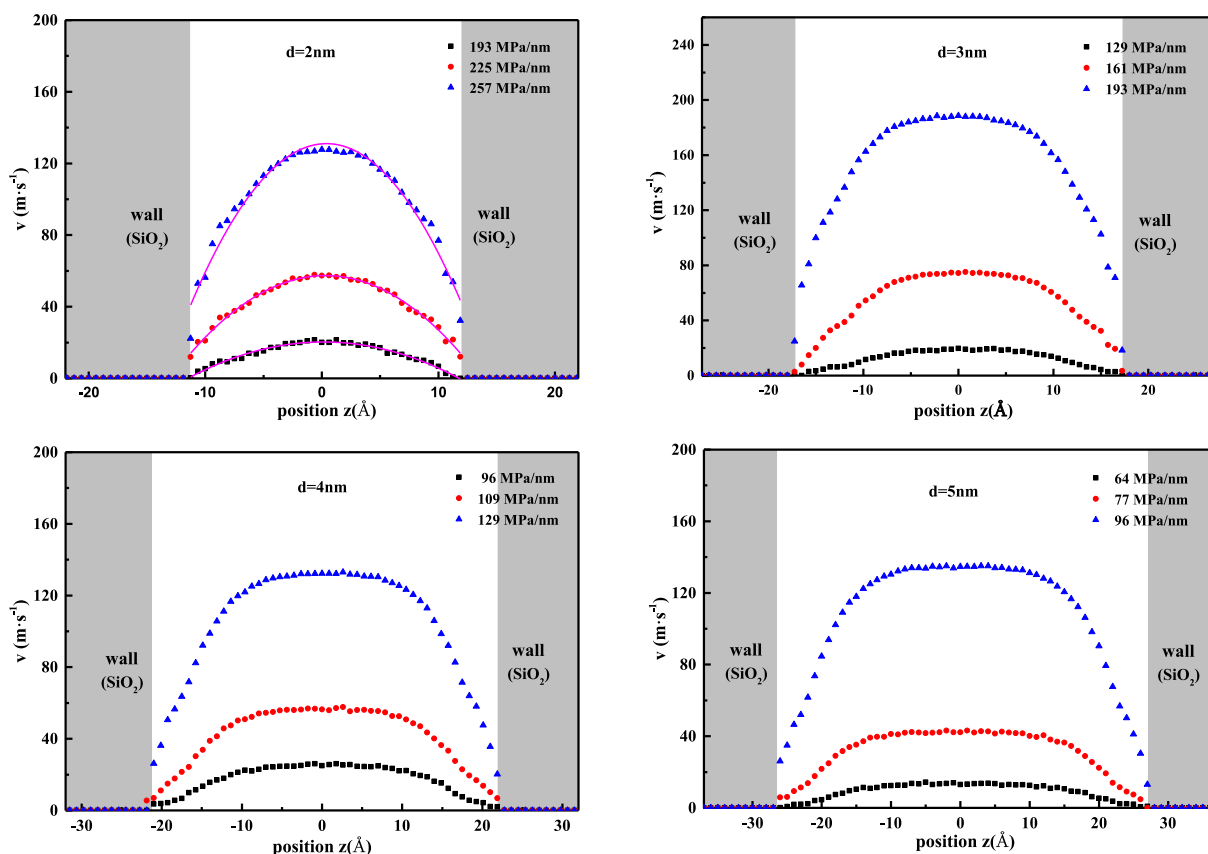


Fig. 13. Velocity profile of [EMIM][BF₄] confined in nanochannels with different heights under various pressure gradients. The line represents the quadratic function of velocity with respect to nanochannel height in z direction.

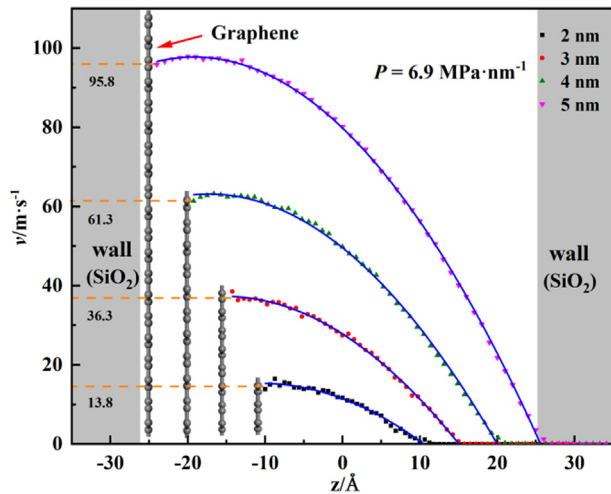


Fig. 14. Velocity of water confined in nanochannel with different heights. The line represents the quadratic function of velocity with respect to nanochannel height in z direction.

sure gradient. Due to the high viscosity of [EMIM][BF₄], the applied pressure gradient needs to be much larger than that of water to make [EMIM][BF₄] flow.

3) Effect of graphene on flow behavior of water and [EMIM][BF₄]

In order to investigate the influence of graphene on the flow behavior of confined water in nanochannel, this section calculates the flow velocity of confined water at pressure gradient of 6.9 MPa·nm⁻¹, as shown in Fig. 14. The inner surface of one side of the silica nanochannel is covered with monolayer graphene. It can be clearly seen that on the side where the graphene is coated, the water has an obvious positive slippage, while on the other side that is not coated with graphene, the negative slippage changes into no slippage. Under the same driving pressure gradient, the slippage increases significantly with the increase of nanochannel height, the maximum velocity is closer to the wall coated graphene and at small nanochannel height. Although water flow presents Poiseuille flow, but the velocity variation curve is a half of parabola. The maximum velocity is higher than the case that no

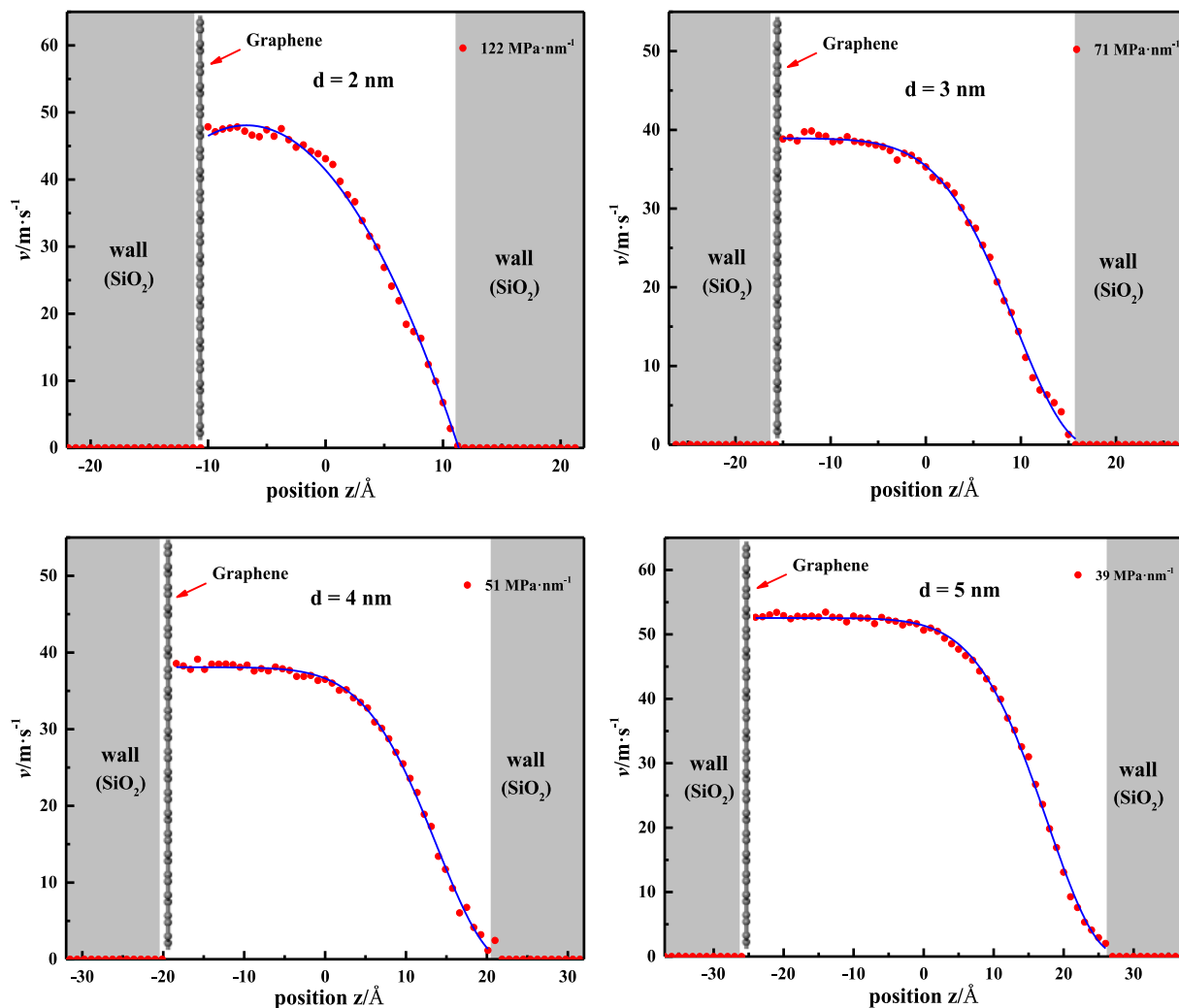


Fig. 15. Flow velocity profile of [EMIM][BF₄] confined in graphene-coated silica nanochannel with various heights. The line in the nanochannel height of 2 nm represents the quadratic function of velocity with respect to nanochannel height in z direction, the others are guide for eyes.

graphene coated, for example, the maximum velocity of water flow in nanochannel whose wall is coated with graphene at $6.9 \text{ MPa}\cdot\text{nm}^{-1}$ is $95.8 \text{ m}\cdot\text{s}^{-1}$ which is higher than that in nanochannel whose wall isn't coated with graphene at $20 \text{ MPa}\cdot\text{nm}^{-1}$, indicates that graphene gives a significant improvement on the flow of water.

The flow of [EMIM][BF₄] in nanochannel coated graphene on one side is shown in Fig. 15. Compared with the case without coated graphene, there is a large positive slippage on the side coated graphene, but no slippage on another side; graphene has a significant effect on improving flow which is indicating by the larger flow velocity under the lower pressure gradient. When nanochannel height is 2 nm, the flow of [EMIM][BF₄] is Poiseuille flow, flow velocity is a half of parabola. As the height of nanochannel increases ($d \geq 3 \text{ nm}$), the flow of [EMIM][BF₄] changes into half partial plunger flow, which is also different from the flow of confined water.

4. Conclusions

In this work, molecular dynamics is used to explore the structure, diffusion behavior and flow characteristics of water and [EMIM][BF₄] confined in asymmetric graphene-coated silica nanochannels aiming to provide information for flow control of molecular and ionic liquid in micro-nano devices.

It is found that combined [EMIM][BF₄] shows more obvious layered structure in silica nanochannel than water, while coating graphene on the surface and reducing the height of nanochannel can enhance the layering effect and uniformize the charge distribution. [EMIM] diffuses faster than [BF₄], coating graphene on wall surface of nanochannel reduces the diffusion coefficients of [EMIM] and [BF₄] as well as their difference. In addition, as the height of nanochannel increases, water flow is always Poiseuille flow, but the negative slippage on the nanochannel wall changes into no slippage. [EMIM][BF₄] exhibits Poiseuille flow in the low height nanochannel with strong confinement, but partial plunger flow when the confinement is weak at a high nanochannel height. Different from water, there is a positive slippage of [EMIM][BF₄] on the nanochannel wall. Coating graphene on one side wall surface of nanochannel can effectively increase the flow velocity of the [EMIM][BF₄] and water, while cause obvious positive slippage on that wall surface, and no slippage on another wall surface.

CRedit authorship contribution statement

Xiangyang Liu: Supervision, Resources, Writing - review & editing. **Xiaotong Zong:** Investigation, Methodology, Software, Writing - original draft. **Sa Xue:** Data curation, Formal analysis. **Hui Liu:** Validation. **Maogang He:** Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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